

Monthly state-level tuberculosis incidence and case detection in Brazil, 2003-2023

Methods

Table of contents

1	Overview	2
1.1	Estimands	2
1.2	The identifying idea	2
2	Data and case definitions	3
2.1	Notifications (SINAN)	3
2.2	Deaths (SIM)	3
2.3	Population (IBGE)	4
3	Model	4
3.1	Natural history and likelihood	4
3.2	Fixed delay distributions	5
3.3	Latent time structure	5
3.4	Time-varying death-reporting adjustment	6
3.5	Priors	6
4	Inference	6
5	Reproducibility and software	7
6	Limitations	7
7	References	7

1 Overview

We estimate monthly tuberculosis (TB) **incidence** and the **case-detection probability** for each of the 27 Brazilian federative units (26 states and the Federal District) over January 2003 to December 2023 (252 monthly time points per unit). Both quantities are latent: neither is observed directly. We infer them from two routinely collected data streams, notifications (SINAN) and deaths (SIM), linked to incidence through a monthly mechanistic natural-history process. The model extends the monthly natural-history model of Chitwood et al. (2025), with the death-reporting adjustment generalised to be time-varying following Chitwood et al. (2021). The model is fit one state at a time. These estimates feed a separate municipality-month panel study of air pollution and TB, so that effects can be reported on the incidence scale rather than the notification scale.

We do not build a municipality-level incidence model: at municipality-month resolution TB deaths are too sparse to separate incidence from detection, and a downscaled outcome would strip the municipality-month variation the panel relies on. Municipality burden mapping (Chitwood et al. 2022) is out of scope here.

1.1 Estimands

For state s and month t we target two quantities:

- $\gamma_{s,t}$: the true incidence rate of symptomatic TB (newly symptomatic cases per person-month), reported per 100,000.
- $\delta_{s,t}$: the case-detection probability, the probability that an incident case that becomes detectable is notified.

These are kept distinct from the observed notification and death counts throughout. Observed counts are modelled as realisations of the latent process; they are never relabelled as incidence.

1.2 The identifying idea

Notifications and deaths are observations generated from incidence through the natural-history process. They identify incidence and the case-detection probability **only because informative priors pin the natural-history parameters** (the survival and case-fatality probabilities and the delay distributions). The death channel is what identifies detection: for a given incidence, more deaths imply that fewer cases were detected and treated. Because this rests on the mortality data, the **death-reporting adjustment is first-order** and is modelled as time-varying over 2003-2023, since the coverage and completeness of death registration in SIM improved substantially over the period. A static adjustment would confound improving registration with falling incidence and rising detection.

2 Data and case definitions

2.1 Notifications (SINAN)

TB notifications are drawn from the Sistema de Informacao de Agravos de Notificacao (SINAN). The monthly notification count for a state is the number of records with treatment-entry type “new case” or “relapse” (TRATAMENTO codes 1 and 2), excluding re-entry after loss to follow-up, transfer, post-mortem notification, and unknown entry types. Records are attributed to the municipality of residence, with the municipality of notification used as a fallback, and then aggregated to the state. Cases are dated by month of diagnosis (DT_DIAG).

From the same records we derive two state-month covariate series:

- **Treatment-outcome fractions.** The closure-status field (SITUA_ENCE) gives, among closed cases, the fraction whose outcome is death ($pMort_{s,t}$) and the fraction lost to follow-up ($pAban_{s,t}$). These enter the mortality model as data, not as estimated parameters.
- **Rapid molecular testing (GeneXpert).** The fraction of notified cases with a recorded rapid molecular test result (TEST_MOLEC), a proxy for diagnostic capacity, used as a covariate in the detection sub-model. The roll-out of Xpert MTB/RIF in Brazil began around 2014.

2.2 Deaths (SIM)

TB deaths are drawn from the Sistema de Informacao sobre Mortalidade (SIM). A death is counted as a TB death if its underlying cause is an active-TB ICD-10 code (A15-A19). The monthly state count uses the same residence-with-occurrence attribution as notifications and is dated by month of death.

From the same all-cause SIM extract we compute the **ill-defined-cause-of-death (IDC) fraction**: the share of all deaths in a state-month coded to ICD-10 Chapter XVIII (R00-R99, “causas mal definidas”). This series is the covariate that drives the time-varying death-reporting adjustment.

Using one all-cause extract for both the TB-death count and the IDC fraction keeps the numerator and denominator on the same attribution rules. State-month completeness is asserted (every state-year must be present, and no state-month may have zero all-cause deaths), so a failed or truncated download cannot be silently treated as zero TB deaths.

2.3 Population (IBGE)

The person-time denominator is the resident population from the Instituto Brasileiro de Geografia e Estatística (IBGE), via the SIDRA system. Annual state-level estimates span the 2000, 2010, and 2022 censuses. We combine the annual intercensal estimates with the 2022 census count (which revised several state populations relative to the projections), giving census priority where they overlap, and interpolate linearly onto monthly mid-points. Population enters the likelihood as a multiplicative person-time offset, not as a covariate.

3 Model

The model is discrete in monthly time and fit separately for each state; the state subscript is suppressed below. To make the fixed-delay convolutions complete for the observed window, the latent series are defined on an extended axis that prepends a pre-window of length equal to the total delay support; only the observed months enter the likelihood.

3.1 Natural history and likelihood

Let λ_t be the per-capita TB infection rate. Incident symptomatic cases arise by convolving infections with the infection-to-symptom delay,

$$\gamma_t = \sum_{i \geq 0} \lambda_{t-i} \phi_i^\lambda,$$

where ϕ^λ is the discretised delay kernel. The symptomatic series γ_t is the incidence estimand. Cases become detectable after a second delay, and a fraction δ_t of detectable cases are notified:

$$\text{Notified}_t = \delta_t \sum_{i \geq 0} \gamma_{t-i} \phi_i^\gamma, \quad \text{Missed}_t = (1 - \delta_t) \sum_{i \geq 0} \gamma_{t-i} \phi_i^\gamma.$$

Deaths arise on two pathways. Among notified cases, a fraction die during or after treatment, governed by the observed treatment-outcome fractions and the conditional probability of death given loss to follow-up. Among missed cases, a fraction die undiagnosed. Both are spread over the detectable-to-death delay ϕ^{mort} :

$$\text{AllDeaths}_t = \sum_{i \geq 0} \left[\text{Notified}_{t-i} (\text{pMort}_{t-i} + \text{pAban}_{t-i} p_{\text{m|aban}}) + \text{Missed}_{t-i} p_{\text{m|undx}} \right] \phi_i^{\text{mort}}.$$

The two observation channels are Poisson, with the population offset pop_t converting per-capita rates to expected counts:

$$\text{SINAN}_t \sim \text{Poisson}(\text{pop}_t \text{Notified}_t), \quad \text{SIM}_t \sim \text{Poisson}(\text{pop}_t \text{AllDeaths}_t \rho_t),$$

where ρ_t is the death-reporting adjustment (Section 3.4). Information from SINAN and SIM about death and loss to follow-up is recorded by notification date, so the treated-death thinning is applied at the notification month, consistent with Chitwood et al. (2025).

3.2 Fixed delay distributions

The three delays are fixed (not estimated), taken from Chitwood et al. (2025) and discretised to per-month probability mass functions, truncated at the stated maxima and renormalised:

Delay	Distribution	Mean (months)	Max (months)
Infection to symptom onset	Weibull(shape 1.75, scale 25)	22.3	60
Symptom onset to detectable	Gamma(shape 10, rate 4)	2.5	7
Detectable to death	Gamma(shape 12, rate 3)	4.0	10

3.3 Latent time structure

Log-incidence and logit-detection are smooth functions of time with three components, rather than a single log-linear trend with one break. For incidence,

$$\log \lambda_t = \alpha_\lambda + \mathbf{B}_t \beta_\lambda + \mathbf{S}_t \omega_\lambda + c_\lambda^{\text{lev}} \text{COVID}_t^{\text{lev}} + c_\lambda^{\text{slo}} \text{COVID}_t^{\text{slo}},$$

and identically for logit δ_t with its own coefficients plus a GeneXpert term $\beta_x \text{Xpert}_t$. Here:

- $\mathbf{B}_t$ is a penalised cubic B-spline basis on coarse (roughly annual-to-biannual) knots, capturing the smooth long-run trend. The spline coefficients carry a first-order random-walk smoothing prior in a non-centred parameterisation.
- $\mathbf{S}_t$ is a Fourier harmonic basis (two harmonics) for the cyclic seasonal effect.
- $\text{COVID}_t^{\text{lev}}$ and $\text{COVID}_t^{\text{slo}}$ encode an explicit level-and-slope shock from April 2020 onward, modelling the pandemic disruption.

We deliberately do **not** include a monthly first-order autoregressive random walk: it overfit and failed to converge in Chitwood et al. (2025) and is excluded here.

3.4 Time-varying death-reporting adjustment

Chitwood et al. (2025) use a static death-reporting adjustment. Over the full 2003-2023 window we instead model it as a logit-linear function of time and the ill-defined-cause fraction, following the structure of Chitwood et al. (2021):

$$\rho_t = \text{logit}^{-1}(\theta_0 + \theta_{\text{time}} \text{year}_t + \theta_{\text{idc}} \text{IDC}_t).$$

This lets improving death registration (a falling IDC fraction and a secular time trend) be separated from changes in true incidence and detection. The biological case-fatality parameters are held constant over time; only the reporting adjustment drifts.

3.5 Priors

The natural-history priors are load bearing and are taken from Chitwood et al. (2025) and Chitwood et al. (2021):

Parameter	Meaning	Prior
$p_{\text{m undx}}$	death given undiagnosed	Beta(113, 87), mean 0.565
$p_{\text{m aban}}$	death given loss to follow-up	Beta(10, 190), mean 0.05
$\alpha_\lambda, \beta_\lambda$	incidence level and trend/COVID	Normal(0, 10)
α_δ , detection coefficients	detection level and trend/COVID	Normal(0, 1)
$\theta_0, \theta_{\text{idc}}$	death-adjustment intercept, IDC coef	Normal(0, 1)
θ_{time}	death-adjustment time trend (per year)	Normal(0, 0.05)

The fixed-delay kernels and the observed treatment-outcome fractions enter as data. The incidence intercept is centred at a realistic per-capita log rate to aid initialisation; this is a numerical centring choice and not an informative constraint.

4 Inference

The model is written in Stan and fit with cmdstanr, one state at a time; a joint hierarchical fit across all 27 states did not converge in Chitwood et al. (2025) and is not used. Each state is sampled with four chains. Convergence is assessed with $\hat{R}$ (target < 1.01), bulk and tail effective sample size, and the number of divergent transitions, computed over the incidence and detection

series and all sampled coefficients, not only a few scalars. We perform a prior predictive check before fitting and a per-state posterior predictive check against observed notifications and deaths after fitting. Convergence on the 252-month series is the hardest part; we address it by reparameterisation (non-centred trend and seasonal terms, informative initialisation of the per-capita intercept) rather than only raising the sampler target acceptance.

The implementation is validated by a recovery test: data simulated from the same generative process with known incidence, detection, COVID, and death-adjustment parameters are fit, and the model is required to recover the latent series (interval coverage and correlation) and the signs of the structural terms. An R implementation of the model’s deterministic core is checked to reproduce the simulator’s expected counts exactly, so the simulator and the Stan model are provably the same data-generating process.

5 Reproducibility and software

All data processing is in R with `data.table`; inference is in Stan via `cmdstanr`. The pipeline is a **targets** directed acyclic graph: raw SINAN, SIM, and IBGE inputs are cached, the state-month panel is assembled, and the 27 per-state fits run as dynamic branches. Every stochastic step is seeded, including the `cmdstanr` fit and the R-side initialisation; paths are relative to the project root; and all artefacts under **outputs/** are regenerable from code and inputs. Package versions are pinned with `renv` and the `cmdstan` version is recorded with each fit.

6 Limitations

TB deaths are defined by underlying cause only, which undercounts deaths where TB is a contributing cause. The GeneXpert and treatment-outcome covariate definitions, the exact ill-defined-cause code set, and the population-denominator reconciliation across the 2010 and 2022 censuses are recorded as decisions pending confirmation. The death-reporting adjustment is identified from the data and the IDC covariate but is only weakly anchored to external coverage estimates; this is the most consequential assumption and is examined in sensitivity analysis.

7 References

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